MIL-C-62207B(AR)
30 June 1981
SUPERSEDING
MIL-C-62207A(AR)
1 March 1976

MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY: 11743667 DIGITAL TO ANALOG CONVERTER

This specification is approved for use by the US Army Armament Research and Development Command (ARRADCOM) and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements and quality assurance provisions for the Circuit Card Assembly: 11743667 Digital to Analog Converter which is the A6 component of the Electronic Unit, 11743131.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-I-45607	Inspection Equipment: Acquisition, Maintenance and Disposition of
MIL-STD-45662	Calibration System Requirement

FSC 1240

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-810	Environmental Test Methods

DRAWINGS

ARRADCOM

11743667	Circuit Card Assembly Digital to Analog Converter

Inspection Equipment

11750202	Automatic Card and Unit, Test Set
11750440	Adapter, Digital/Analog Converter
11821457	Fixture Shock, Vertical and Horizontal Plane
11821458	Shock Adapter, Plate Vib Horiz - Vert Plane/ Shock-Z
11821520	Automatic Test Assembly, Central Processor Unit
11821647	Vibration/Shock Test Fixture, Electron Units FAT
11821861	Test Fixture - Auto Card Test Set, Electrical Unit FAT

Packaging Data Sheet

P11743667	Packaging of Circuit Card Assembly: 11743667 Digital to Analog Converter

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Fabrication. The digital to analog converter circuit card assembly, herein referred to as the assembly, shall be manufactured in accordance with Drawing 11743667 and drawings pertaining thereto and, when assembled, shall meet the requirements of this specification. (See 4.5.1)

3.1.1 Function. The assembly shall provide output voltages as specified from designated input conditions.

3.1.2 General specifications. The following provisions of MIL-P-13926 apply: (See 4.5.1)

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishes
d. Part identification and marking
e. Workmanship

3.1.3 Ambient conditions. Standard ambient conditions shall be as follows:

a. Temperature	73° ± 18°F
b. Relative humidity	50 percent ± 30 percent
c. Atmospheric pressure	28.5 + 2.0 -3.2 in. Hg.

3.2 First article. When specified (see 6.2), the contractor shall furnish sample units for first article inspection and approval (see 4.4 and 6.2).

3.3 Performance. Unless otherwise specified, the assembly shall meet the performance requirements specified herein under standard ambient contion of 3.1.3.

3.3.1 Output voltage. with the laser range signal (item 3.2) of table I input conditions established for each item of table II in the item number order, and with the logical one strobe signal (item 3.1) of table I applied at completion of establishment of laser range signal input conditions for each item, the corresponding output voltage at pin A31 shall be as specified in table II. (See 4.6.2.1)

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TABLE I. Loads, power and signals.

Item	Conditions	Characteristics	Connections
1.0	Loads		
1.1	Resistor	33.0 kilohms ±5 percent, 1/4 watt	Connected between pins A31 and A27
2.0	Power sources		
2.1	15.00 Vdc	Tolerance: ±0.75 Vdc Ripple: 25 mV p-p max Current: 150 mA	Applied between pins B28 (+) and A27 (-)
2.2	5.00 Vdc	Tolerance: ±0.2 Vdc Ripple: 25 mV p-p max Current: 200 mA	Applied between pin B22 (+) and pins A36 (-) and B16 (-)
2.3	-15.00 Vdc	Tolerance: ±0.75 Vdc Ripple: 25 mV p-p max Current: 150 mA	Applied between pins A27 (-) and B26 (+)
2.4	5.0000 Vrms	Tolerance: ±0.0020 Vrms 1/ 400 ± 4.0 Hertz	Applied between pins B30 and A27 (rtn)
	or +5.0000 Vdc and successively -5.0000 Vdc	Tolerance: ±0.0020 Vdc 1/ Ripple: 100 micro V p-p max	
3.0	Signals		
3.1	Strobe	Logical one: 4.0 ± 1.0 V (Source Impedance: 2 kilohms ± 5%) Logical zero: 0.2 ± 0.2 V	As specified, applied between pin A17 and pins A36 (rtn) and B16 (rtn)
3.2	Laser range	Logical one: 4.0 ± 1.0 V (Source Impedance: 2 kilohms ±5%) Logical zero: 0.2 ± 0.2 V	Applied as specified

1/ The tolerance on the reference voltages may be increased to ±0.05 V when ratio measurement method is used.

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TABLE II. Output voltages.

Item	Laser Range Signal Input Conditions at Pins 1/.								
	B2 (4000)	A3 (2000)	B4 (1000)	A5 (300)	B6 (400)	A7 (200)	B8 (100)	A9 (80)	B10 (40)
1	1	1	1	1	1	1	1	1	1
2	1	1	1	1	1	1	1	1	1
3	1	1	1	1	1	1	1	1	1
4	1	1	1	1	1	1	1	1	1
5	1	1	1	1	1	1	1	1	0
6	1	1	1	1	1	1	1	0	1
7	1	1	1	1	1	1	0	1	1
8	1	1	1	1	1	0	1	1	1
9	1	1	1	1	0	1	1	1	1
10	1	1	1	0	1	1	1	1	1
11	1	1	0	1	1	1	1	1	1
12	1	0	1	1	1	1	1	1	1
13	0	1	1	1	1	1	1	1	1
14	1	1	0	0	1	1	1	1	1
15	0	1	1	1	0	0	0	1	1
16	0	0	0	0	0	0	0	0	0

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TABLE II. Output voltages. - Continued

Item	Laser Range Signal Input Conditions at Pins 1/			Output Voltage at Pin A31 2/	OR Ratio 3/
	A11 (20)	B12 (10)	A13 (5)		
1	1	1	1	0.0000 ± 0.0100 for DC. Less than 0.0020 for AC.	- -
2	1	1	0	0.0053 ± 0.0020	0.00106 ± 0.00040
3	1	0	1	0.0106 ± 0.0020	0.00213 ± 0.00040
4	0	1	1	0.0213 ± 0.0020	0.00426 ± 0.00040
5	1	1	1	0.0426 ± 0.0020	0.00852 ± 0.00040
6	1	1	1	0.0851 ± 0.0020	0.01702 ± 0.00040
7	1	1	1	0.1064 ± 0.0020	0.02128 ± 0.00040
8	1	1	1	0.2128 ± 0.0020	0.04256 ± 0.00040
9	1	1	1	0.4256 ± 0.0020	0.08512 ± 0.00040
10	1	1	1	0.8512 ± 0.0020	0.17024 ± 0.00040
11	1	1	1	1.0640 ± 0.0040	0.21280 ± 0.00080
12	1	1	1	2.1280 ± 0.0060	0.42560 ± 0.00120
13	1	1	1	4.2560 ± 0.0080	0.85120 ± 0.00160
14	0	1	0	1.9420 ± 0.0040	0.38840 ± 0.00080
15	1	1	1	5.0000 ± 0.0080	1.00000 ± 0.00160
16	0	0	0	1.70 ± 0.05	0.34 ± 0.05

1/ Input signal condition legend:
1 = logical one strobe signal, item 3.1 of table I (source impedance: 2 kilohms ±5%)
0 = logical zero strobe signal, item 3.1 of table I

2/ Output voltage in V rms or (+) Vdc and successively (-) Vdc. For DC output voltage, Vdc = $\frac{(1 + \text{Vdc out} + 1 - \text{Vdc out } 1)}{2}$

3/ Ratio: $\frac{\text{Output voltage at pin A31 (Vat) or}}{\text{Ref. voltage at pin B30 (Vat)}}$

$$\frac{(1 + \text{Vdc Out} + 1 - \text{Vdc Out})}{2} \text{ (Average)}$$

when this method is used to verify the output, the tolerance on the ref. voltages(item 2.4 in table I) may be 1%.

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3.4 Environmental.

3.4.1 Shock. The assembly shall be capable of operating as specified hesrein after exposure under the conditions of 3.1.3 to three half sine wave shock pulses of 40 ± 4 gravity units (g) for a duration of 18 ± 3 milliseconds (msec) applied in each direction along three mutually perpendicular axes. In addition, the assembly shall be capable of withstanding three half sine wave shock pulses of 100.0 ± 10.0 g for a duration of 1.5 ± 0.2 msec applied in each direction along three mutually perpendicular axes. (See 4.6.3.1)

3.4.2 Vibration. The assembly shall be capable of operating as specified herein after exposure under the conditions of 391.3 to the vibration Profile of figure 1. Duration of exposure shall be not less than 80 minutes in each of three mutually perpendicular axes. (See 4.6.3.2)

3.4.3 Operating temperature. The assembly shall be capable of operating as specified herein over the operating temperature range of -25°F to +140°F. (See 4.6.3.3, 4.6.3.4 and 4.6.3.5)

3.4.4 Storage temperature. The assembly shall be capable of operating as specified herein after exposure to storage temperatures ranging from -65°F to +160°F. (See 4.6.3.3, 4.6.3.4 and 4.6.3.5)

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herin. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unlcess disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.4).
2. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the quality assurance provisions of MIL-F-13926 and the conditions of 3.1.3.

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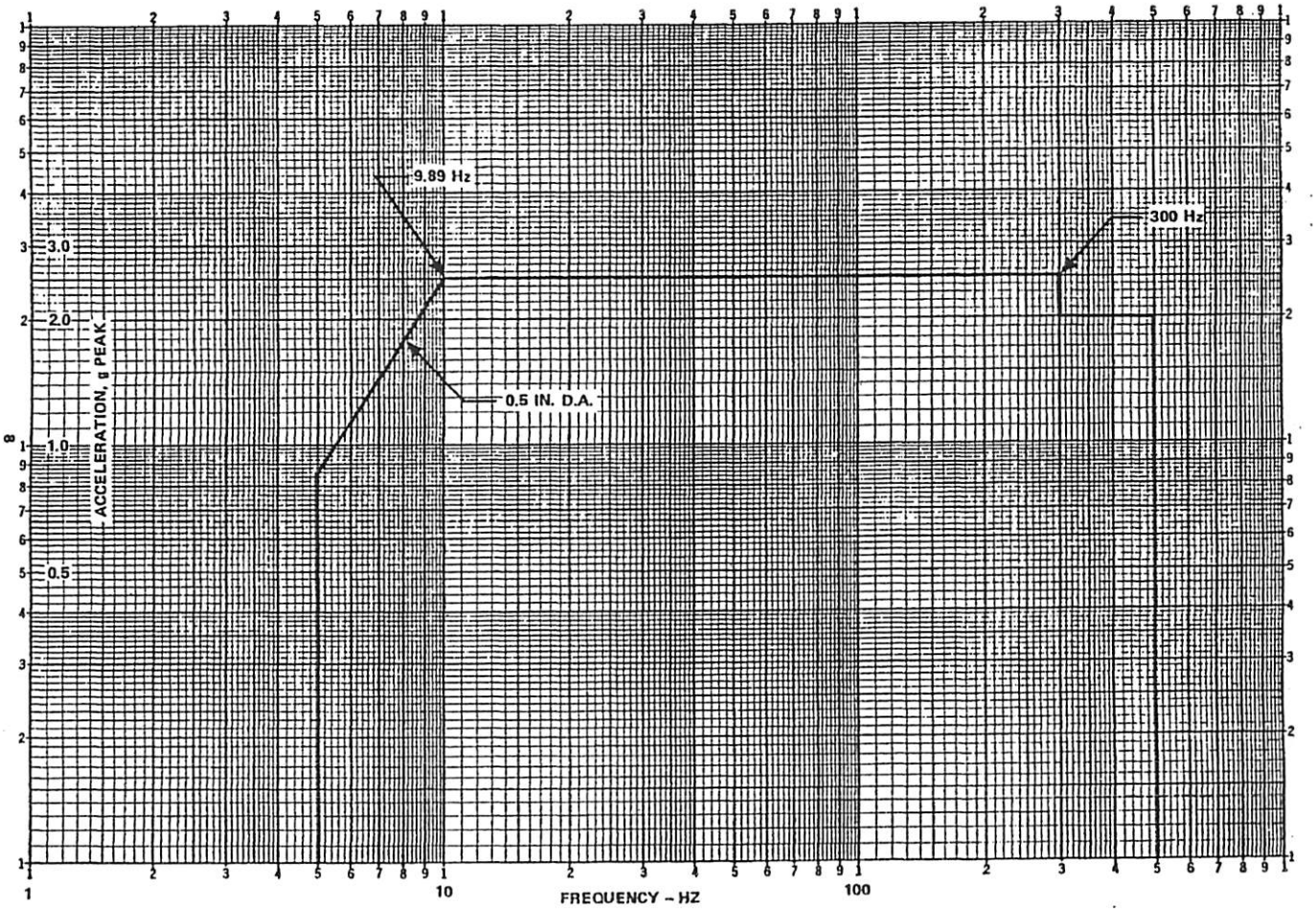


FIGURE 1. Vibration profile.

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4.4 First article. The requirement for first article approval and the designation of responsibility for first article inspection to either the Government or the contractor shall be specified in the contract.

4.4.1 Sample. The first article sample shall be three assemblies selected at random by the Government representative from the first fifteen production assemblies.

4.4.2 Inspection. The sample shall be subjected to all the inspections specified in tables III, IV and V.

4.4.3 Failure. Failure of any assembly to meet any requirement shall be cause for refusal to grant first article approval. The Government reserves the right to terminate first article inspection upon any failure of any assembly to comply with any stated requirement.

4.4.4 Responsiblity. The contractor, whether or not responsible, shall inspect the sample for conformance to all contractual requirements and shall submit a record-of this inspection with the sample and certificates of conformance for materials. The Government reserves the right to witness inspections performed by the contractor.

TABLE III. Performance tests.

Item	Characteristic	Requirement	Test Procedure
101	Output voltage	3.3.1	4.6.2.1

TABLE IV. General tests.

Item	Characteristic	Requirement	Test Procedure
301	Fabrication	3.1	Applicable drawings - Visual
302	General specifications	3.1.2	MIL-F-13926 - Visual

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TABLE V. Environmental tests.

Item	Characteristic	Requirement	Test Procedure
303	Shock	3.4.1	4.6.3.1
304	Vibration	3.4.2	4.6.3.2
305	High Temperature	3.4.3 and 3.4.4	4.6.3.3
306	Low Temperature	3.4.3 and 3.4.4	4.6.3.4
307	Temperature Cycling	3.4.3 and 3.4.4	4.6.3.5

4.5 Quality conformance inspections

4.5.1 Procurement conditions. The following inspection plans shall apply where the assembly is procured for use as a:

Use	Plan
a. Logistics spare assembly apart from the next higher assembly	A and C
b. Component assembly of the next higher assembly	B and C

4.5.2 Inspection plan A.

4.5.2.1 General sample and tests. One assembly, as a control sample, shall be selected at random by the Government representative from each 100 assemblies produced and shall be subjected to all the test in table IV.

4.5.2.2 Environmental sample and tests. Three assemblies, as a control sample, shall be selected at random by the Government representative from each 50 assemblies produced or from each month's production, whichever occurs first. All the tests in table V shall be applied separately to each assembly in the sample.

4.5.2.3 Acceptance. Where any one assembly of either sample fails to meet any specified requirement, the lot shall be rejected. Rejected lots shall be subject to the provisions of MIL-STD-105.

4.5.3 Inspection plan B.

4.5.3.1 Sample and tests. The sample shall be selected by the Government representative in accordance with the provisions of MIL-STD-105. All the tests in table IV and only the test of item 307 in table V shall be applied.

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4.5.3.2 Acceptance. Acceptance and rejection shall be in accordance with MIL-STD-105.

4.5.4 Inspection plan C.

4.5.4.1 Sample and tests. Each assembly in every lot shall be subjected to all the tests in table III and shall be examined visually for completeness, improper assembly and evidence of poor workmanship.

4.5.4.2 Acceptance. Where any one assembly fails to meet any specified requirement, the defective assembly shall be removed from the lot and resubmitted only after all defects have been corrected.

4.5.5 Inspection equipment. Unless otherwise specified in the contract, the contractor shall supply, maintain and calibrate inspection equipment in accordance with the provisions of MIL-I-45607 and MIL-STD-45662

4.5.6 Packaging inspection. The sampling and inspection of the preservation - packaging, packing and container marking shall be in accordance with the provisions of packaging data sheet P11743667.

4.6 Methods of inspection.

4.6.1 Test equipment. Unless the assembly is procured apart from the next higher assembly, the test equipment in table VI shall be used where specified to perform the required test.

TABLE VI. Test equipment.

Item	Part Number	Nomenclature
1	11750202	Automatic Card and Unit, Test Set
2	11750440	Adapter, Digital/Analog Converter
3	11821457	Fixture Shock, Vertical and Horizontal Plane
4	11821458	Shock Adapter, Plate Vib Horiz-Vert Plane/Shock-Z
5	11821520	Automatic Test Assembly, Central Processor Unit
6	11821647	Vibration/Shock Test Fixture, Electron Units FAT
7	11821861	Test Fixture - Auto Card Test Set, Electrical Unit FAT

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4.6.2 Performance tests.

4.6.2.1 Output voltage. Use items 1, 2 and 5 of table VI or suitable equipment to determine compliance with 3.3.1.

4.6.3 Environmental tests.

4.6.3.1 Shock.

4.6.3.1.1 Basic design. Use items 3, 4 and 6 of table VI or mount the assembly in a suitable shock fixture and subject it to the shock test specified in MIL-STD-810, method 516.2, procedure I, figure 516.2-2. Apply three half sine wave shock pulses in each direction along the three axes. Peak amplitude shall be 40 ±4 g with a time duration of 18 ±3 msec measured at the 10 percent amplitude points. At the conlusion of this test, subject the assembly to the performance tests of table III.

4.6.3.1.2 High intensity. Use items 3, 4 and 6 of table VI or mount the assembly in a suitable shock fixture and subject it to the shock test specified in MIL-STD-810, method 516.2, procedure IV, figure 516.2-2. Apply three half sine wave shock pulses in each direction along the three axes. Peak amplitude shall be 100.0 ±10.0 g with a time duration of 1.5 ±0.2 msec measured at the 10 percent amplitude points. At the conclusion of this test, subject the assembly to the performance tests of table III.

4.6.3.2 Vibration. Vibration A as specified in 4.6.3.2.1 shall be conducted only for first article inspection. Vibration B as specified in 4.6.3.2.2 shall be conducted only for quality conformance inspection.

4.6.3.2.1 Vibration A. Use items 4 and 6 of table VI or mount the assembly in a suitable vibration fixture and subject the assembly to the vibration test specified in MIL-STD-810, method 514.2, procedure VIII, except that the test level shall be the vibration curve shown in figure 1 and duration of exposure shall be not less than 80 minutes per axis. At the conclusion of this test, subject the assembly to the performance tests of table III.

4.6.3.2.2 Vibration B. This test shall be conducted as in 4.6.3.2.1 except that the period of vibration shall be not less than 15 minutes in each axis.

4.6.3.3 High temperature. Use item 7 of table VI or suitable equipment and subject the assembly to the high temperature test specified in MIL-STD-810 method 501.1, procedure I. The highest operating temperature shall be 125°F. The performance tests of table III shall be applied.

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4.6.3.4 Low temperature. Use item 7 of table VI or suitable equipment and subject the assembly to the low temperature test specified in MIL-STD-810, method 502.1, procedure I, The storage temperature shall be -65°F. The lowest operating temperature shall be -25°F. The performance tests of table III shall be applied.

4.6.3.5 Temperature cycling. With power and loads of table I applied, subject the assembly to the temperature profile specified in figure 2. At the conclusion of this test, subject the assembly to the performance tests of table III.

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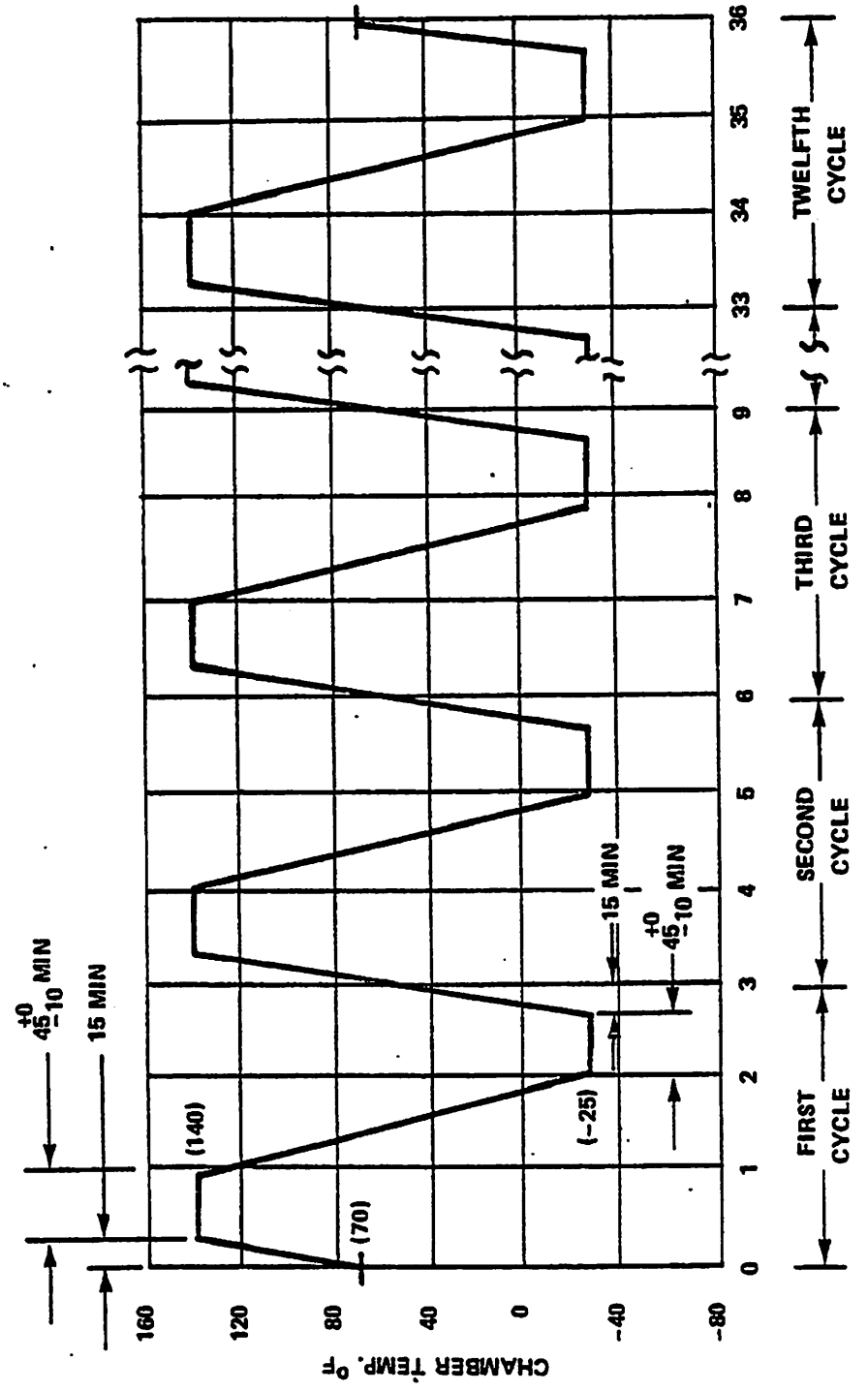


FIGURE 2. Temperature cycling.

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5. PACKAGING

5.1 Packaging, packing and marking. Packaging, packing and marking shall be in accordance with packaging data sheet P11743667. The level of protection shall be as specified in the procurement document. (See 4.5.6)

6. NOTES

6.1 Intended use. The Digital to Analog Converter Circuit Card Assembly is a functional part of the Electronic Unit, intended for use in the Laser Fire Control Rangefinder (AN/VVG-2). The function of the assembly is to provide a laser range signal which is used by the computer system.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Special tests and examination as required, the type of inspection required and the activity responsible for testing and its facility.

c. Applicable packaging data sheet number (see 5.1).

d. Selection of applicable levels of preservation, packaging and packing.

e. Applicable stock number.

f. First article inspection requirement.

6.3 Previous revisions of MIL-C-62207B was for the US Army.Armament. Research and Development Command's internal use only, and were not forwarded to the Naval Publications and Form Center for publication.

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